UNIVERSIDADE DE SÃO PAULO
Instituto de Ciências Matemáticas e de Computação

Asymptotics of Partition Functions of Coulomb Gases

João Victor Alcântara Pimenta

Dissertação de Mestrado do Programa de Pós-Graduação em Matemática (PPG-Mat)

ICMC USP
SÃO CARLOS

SERVIÇO DE PÓS-GRADUAÇÃO DO ICMC-USP

Data de Depósito:

Assinatura: ______________________

João Victor Alcântara Pimenta

Asymptotics of Partition Functions of Coulomb Gases

Master dissertation submitted to the Institute of Mathematics and Computer Sciences – ICMC-USP, in partial fulfillment of the requirements for the degree of the Master Program in Mathematics. *EXAMINATION BOARD PRESENTATION COPY*

Concentration Area: Mathematics

Advisor: Prof. Dr. Guilherme Silva

USP – São Carlos
July de 2026

Ficha catalográfica elaborada pela Biblioteca Prof. Achille Bassi
e Seção Técnica de Informática, ICMC/USP,
com os dados fornecidos pelo(a) autor(a)

Antonelli, Humberto Lidio
A634m Modelo de teses e dissertações em LaTeX do ICMC / Humberto Lidio Antonelli; orientadora Renata Pontin de Mattos Fortes. -- São Carlos, 2017.
79 p.

Tese (Doutorado - Programa de Pós-Graduação em Ciências de Computação e Matemática Computacional) -- Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, 2017.

1. Modelo. 2. Monografia de qualificação. 3. Dissertação. 4. Tese. 5. Latex. I. Fortes, Renata Pontin de Mattos, orient. II. Título.

João Victor Alcântara Pimenta

Asymptotics of Partition Functions of Coulomb Gases

Master dissertation submitted to the Institute of Mathematics and Computer Sciences – ICMC-USP, in partial fulfillment of the requirements for the degree of the Master Program in Mathematics. *EXAMINATION BOARD PRESENTATION COPY*

Concentration Area: Mathematics

Advisor: Prof. Dr. Guilherme Silva

USP – São Carlos
July de 2026

"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.
Le monde est désordre.
L'équilibre n'est pas la règle, c'est l'exception.
Et je faisais le serment de travailler pour l'ordre et l'équilibre.
Que de serments! Tu vas sourire."
G.Duhamel, Maitres, 1937

ACKNOWLEDGEMENTS

For all of those who are or were:
Your courage, patience, daring, and caring has brought me here.
To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.
For all of those who will be:
To promise that I can for you as it was done for me.

"João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito."
Te amo muito, Papai.

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas de Funções Partição em Gases de Coulomb**. 2026. 53 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

As matrizes aleatórias podem ser vistas de duas formas: como uma função mensurável de um espaço de probabilidade num conjunto de matrizes ou, mais concretamente, como uma matriz cujas entradas são variáveis aleatórias com alguma distribuição conjunta. Um conjunto de tais matrizes com uma dada distribuição arbitrária é designado por ensemble. Para ensembles de interesse, pode-se estudar estatísticas como autovetores e distribuições de autovalores. As simetrias do conjunto escolhido irão refletir sobre quais modelos tais estatísticas podem representar. Por exemplo, as distâncias dos valores característicos de uma matriz simétrica com coeficientes aleatórios modelam as distâncias entre níveis sucessivos de energias de núcleos atômicos pesados. Mais interessante ainda, existe uma certa universalidade nas estatísticas encontradas entre classes de ensembles, sugerindo a relevância da teoria no estudo da universalidade nos fenómenos modelados em geral. Particularmente importante é o estudo de tais matrizes quando estas crescem infinitamente, considerando assintoticamente o seu comportamento estatístico. O nosso foco é uma técnica para derivar assintóticas para a chamada função partição de alguns ensembles interessantes de matrizes aleatórias.

Palavras-chave: Teoria de Matrizes Aleatórias, Ensembles de Matrizes Planares, Assintótica.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 53 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set of such matrices with a given arbitrary distribution is called an ensemble. One can, for ensembles of interest, study statistic such as the eigenvectors and eigenvalue distributions. The symmetries of chosen set will reflect on what models such statistics can model. For example, distances of the characteristic values of a symmetric matrix with random coefficients model distances between successive levels of energies of heavy nuclei. More interestingly, there is some universality in the statistics found among classes of ensembles, suggesting the relevance of the theory in studying the universality in general modeled phenomena. Particularly important is the study of such matrices when they grow infinitely large, taking asymptotically its statistical behavior. We focus on a technique for deriving asymptotics for the so-called partition function of some interesting ensembles of random matrices.

Keywords: Random Matrix Theory, Planar Matrix Ensembles, Asymptotics.

LIST OF FIGURES

CONTENTS

CHAPTER

1

INTRODUCTION

CHAPTER

2

RANDOM MATRIX THEORY

There are many instances where one might want to describe the behavior of a set of random points in time or in space. This could come from an investigation of systems of diffusing particles or occurrences in time of some random phenomena. Point processes (PP) are models to describe such random objects: It describes the number of occurrence of events in a given time-frame or, alternatively, the number of points in regions of some space. For this work, we mainly focus on the latter application: density of points in regions of space; especially density of eigenvalues calculated from *random matrices*.

Random matrices can be defined in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such random matrix. We will be interested in these cases, which we call *Invariant Ensembles*. Moreover, we will be particularly interested in a subset of ensembles in which this density is expressed as a *Kernel* - a matricial object that contains the statistical characterization of the system. This is called determinantal structure and will lead us to the concept of Determinantal Point Processes (DPP).

What follows is our best attempt in introducing the concepts in Random Matrix Theory (RMT) that are pertinent to the scope of this work and of general interest, albeit inside the limitations of time, effort and experience. We wrote the proofs that were within the theoretical reach of the chapter and that further elucidated the concepts discussed. More technical results were only stated, with some indication on how one would go about the proof. Definitions and some structure on RMT, in the chapter that follows, are based on the well written Thesis by Rahman (2023). Furthermore, definitions and some ideas for point processes and determinantal point processes follow very closely the well structured works of Serfozo (1990), Johansson (2006) and Girotti (2014).

2.1 Random Matrix

The idea of *random matrices* - matrices with random entries - is a core concept in this work and the main theme of this section. If a matrix *models a system*, we think of each realization of this random object as one state the system can assume. Borrowing the concept from physics, an *ensemble* is thought of as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. To consider an ensemble of random matrices is to associate each state to a probability. More concretely, in the context of Random Matrix Theory (RMT) we have the following definition.

Definition 2.1. *A* ***random matrix ensemble*** $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ *is a set of matrices* $\mathcal{S}$ *of fixed size equipped with probability density function (p.d.f.)* $\mathcal{P} : \mathcal{S} \to [0, \infty)$. *If we say that a matrix* $\hat{X}$ *is drawn from the random ensemble* $\mathcal{E}$, *we are saying that it is a random variable assuming values in* $\mathcal{S}$ *with p.d.f.* $\mathcal{P}$.

The construction of many ensembles in RMT is done by applying some constraint to the set of real, complex or quaternionic matrices. The constraints used are typically those requiring the elements of the set $\mathcal{S}$ to exhibit features such as orthogonality, unitarity, (skew-) symmetry, Hermiticity, invariance under complex conjugations or some reasonable combination of them.

Example 2.1. The $m \times n$ real, complex, and quaternionic *Ginibre Ensembles* are the set of n by m matrices with real, complex and quaternionic entries with probability density function

$$\mathcal{P}^{(Gin)}(\hat{G}) = \frac{1}{\mathcal{Z}^{(Gin)}_{n,m,\beta}} \exp\left(-\sqrt{nm}\,\mathrm{Tr}(\hat{G}^*\hat{G})\right) = \frac{1}{\mathcal{Z}^{(Gin)}_{n,m,\beta}} \prod_{i=1}^{m}\prod_{j=1}^{n} \exp\left(-|G_{i,j}|^2\right),$$

defined with respect to the Lebesgue measure

$$\mathrm{d}\hat{G} = \prod_{i=1}^{m}\prod_{j=1}^{n}\prod_{s=1}^{\beta} \mathrm{d}G^{(s)}_{i,j}.$$

The coefficient β is called the Dyson index and counts the generic number of real components of each matrix entry $G_{i,j}$, so that $\beta = 1, 2, 4$ refers to real, complex and quaternionic entries. These are the Ginibre orthogonal, unitary and symplectic ensembles.

Normally, when constructing an ensemble of random matrices, one is trying to model some type of system or behavior - be that for a physical or mathematical reason. Historically, many ensembles were constructed with Hermitian matrices to model large complex systems such as heavy nuclei, interacting particle gases on the line and other correlated systems. Although you can construct ensembles in many ways, there are two approaches that we will take interest on in this work. They are referred here as the *universality* and the *invariance* methods. The first approach is mainly considered for its contextual and historical importance, however, most results presented in the following chapters are constructed with the second approach in mind.

The universality method consists of assuming very little of the system and focusing on universal results, as suggested by its name. Its origins refer back to the works of Wigner in heavy nuclei as described in Chapter 1. The intent was to capture statistical information on the behavior of complex systems without the need to encode the finer details of the interactions, only demanding symmetries or general features to be displayed. Surprisingly, doing so still allows you to characterize aspects that turn out to be common across a large range of random objects. More precisely, consider the definition below.

Definition 2.2. *Let every $\hat{Y}$ be a matrix $\hat{Y} = \{Y_N\}_{i,j}$ such that $Y_{i,j} = Y^*_{j,i}$ satisfying the following properties: $\{Y_{i,j}\}_{1\le i\le j\le n}$ are independent; $\{Y_{i,j}\}_{i\ne j}$ are identically distributed as also are the $\{Y_{i,i}\}$; $\mathbb{E}(Y_{i,j}) = 0$ and $\mathbb{E}(Y^k_{i,j}) < \infty$, i.e. $r_k = \max\left(\mathbb{E}(Y^k_{i,j}), \mathbb{E}(Y^k_{i,j})\right) < \infty$. Set $r_2 < \infty$ with $\mathbb{E}(Y^2_{1,2}) > 0$. Then, we say $\hat{Y}$ is a* ***Wigner matrix****. Furthermore, if $\hat{X}_n = n^{-\frac{1}{2}}\hat{Y}_n$, $\hat{X}$ is what we call a* ***normalized Wigner matrix****.*

Example 2.2. Let $\hat{G}$ be drawn from the $n \times n$ real, complex or quaternionic Ginibre Ensemble introduced in Example 2.1. Then, the random matrix $\hat{H} = \frac{1}{2}(\hat{G}^* + \hat{G})$ represents an element of the corresponding real, complex or quaternionic *Gaussian ensemble* - also known as *Hermite*. The probability density function for such ensembles are then given by

$$\mathcal{P}^{(G)}(\hat{H}) = \frac{1}{\mathcal{Z}^G_{n,\beta}} \exp\left\{-n \operatorname{Tr}\left(\hat{H}^2\right)\right\}.$$

Although thinking of the Gaussian ensembles as a symmetrization of the Ginibre ensemble can be useful, one can also construct these ensembles thinking of it as a Wigner matrix. Let $\hat{M}$ have its upper-triangular entries $\{M_{i,j}\}_{i<j}$ independently and identically distributed as standard (real, complex or quaternionic) Gaussian random variables. Let also its diagonal entries $\{M_{i,j}\}_{i=j}$ be independently identically distributed as standard real Gaussian random variables with half the variance. Finally, let the matrix be self-adjoint, i.e. $\{M_{i,j}\} = \{M^*_{j,i}\}$ for $i \ne j$ and the eigenvalues scaled by $1/\sqrt{n}$. Such matrix $\hat{M}$ has distribution

$$\mathcal{P}^{(G)}(\hat{H}) = \prod_{i=1}^{n} \frac{\mathrm{e}^{-nh^2_{i,j}}}{\sqrt{\pi}} \prod_{i<j}^{n} \frac{\mathrm{e}^{-2nh^2_{i,i}}}{\sqrt{\pi/2}} = 2^{n(n+1)/4}\pi^{n(n+1)/4} \exp\left\{-n \operatorname{Tr}\left(\hat{H}^2\right)\right\}.$$

Representatives from the Gaussian ensembles are an instance of Wigner matrices as entries are Gaussian random variables satisfying the requirements in Definition 2.2.

The advantages of this approach for constructing ensembles becomes clearer when considering that many convergence results hold for independently and identically distributed centered random variables. We will present, without proof, a theorem that sits at the center of this ideology: It roughly states that the typical eigenvalue configuration, a set of random points, converges to a distribution called *Wigner Semi-Circle* as we increase the size of the matrix. To state it, we first introduce an *empirical spectral measure* of the eigenvalues.

Let $\hat{M}$ be a Wigner matrix, by the spectral theorem $\hat{M}$ has n real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then, define $\delta_{\hat{M}}$, the *empirical spectral measure*, as

$$\delta_{\hat{M}} := \frac{1}{n} \sum_{k=1}^{n} \delta_{\lambda_k}, \tag{2.1}$$

where $\delta_{\lambda_k}(A) \equiv \mathbb{1}_A(\lambda_k)$ is the Dirac delta function. Let $\mathcal{E}$ denote the class of Borel sets and $\mathcal{B}$ the class of bounded Borel sets, both on the real line. One can notice that this function ξ defines a *random counting measure* in the usual sense, i.e. a mapping $f : \mathcal{E} \to \{0, 1, \cdots, \infty\}$ such that $f(\emptyset) = 0$, and $f(B) < \infty$ for any $B \in \mathcal{B}$ and that, for disjoint sets $B_1, B_2, \cdots \in \mathcal{B}$ it is such that $f\left(\bigcup_k B_k\right) = \sum_k f(B_k)$. For Wigner matrices, it can be shown that this discrete measure converges to a continuous measure when we take the limit on the size n of the matrices - when considering a large number of eigenvalues. The is the result proved by Wigner (1958).

Theorem 2.1. *(Wigner's Semicircle Law) The empirical spectral measure $\delta_{\hat{M}}$ of Wigner matrices converges weakly to the Semi-Circle distribution $\mu^{(SC)}$ defined by*

$$\mu^{(SC)}(\mathrm{d}x) := \begin{cases} \frac{1}{\pi R^2}\sqrt{R^2 - x^2}\mathrm{d}x, & |x| \leq R, \\ 0, & |x| > R. \end{cases}$$

where R depends only on the variance of entries of the matrix.

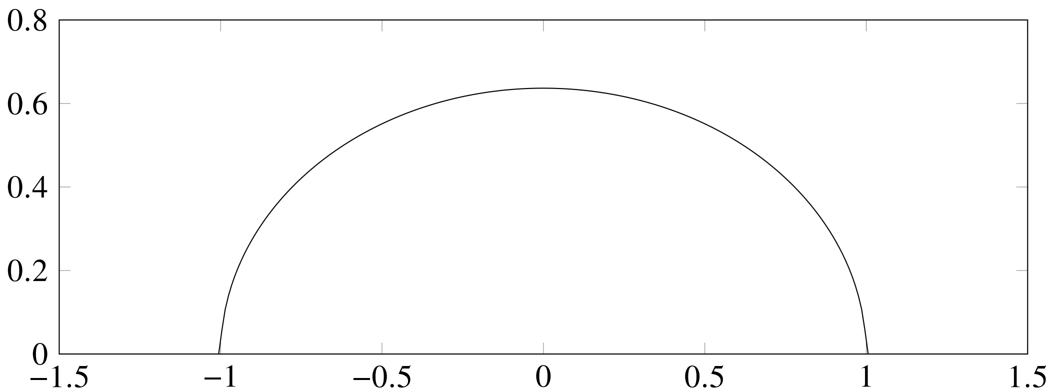

Figure 1 – Plot of the Wigner Semi-Circle distribution normalized such that $\operatorname{supp} \mu^{(SC)} = [-1, 1]$.

This is an instance of what is called *universality* in RMT, hence the approach's name. This will be better formalized when we discuss the *determinantal point processes* in section 2.2. For now, universality refers to the phenomena where different ensembles have the same large limit statistic, e.g. the distribution of eigenvalues. We know that Wigner matrices share the same limit distribution for eigenvalues; other classes of universality may share other statistics: Eigenvalue spacing and maximum eigenvalue distribution are some of them.

Example 2.3. (Numerics on the Semicircle Law) Take the Gaussian Orthogonal (GOE), Unitary (GUE) and Symplectic (GSE) Ensembles. By Theorem 2.1, we expect to see, as $n \longrightarrow \infty$, the empirical eigenvalue measure $\delta_{\hat{M}}$ approximating the Semi-Circle distribution μ^{SC}. We indicate such convergence in the simulations below.

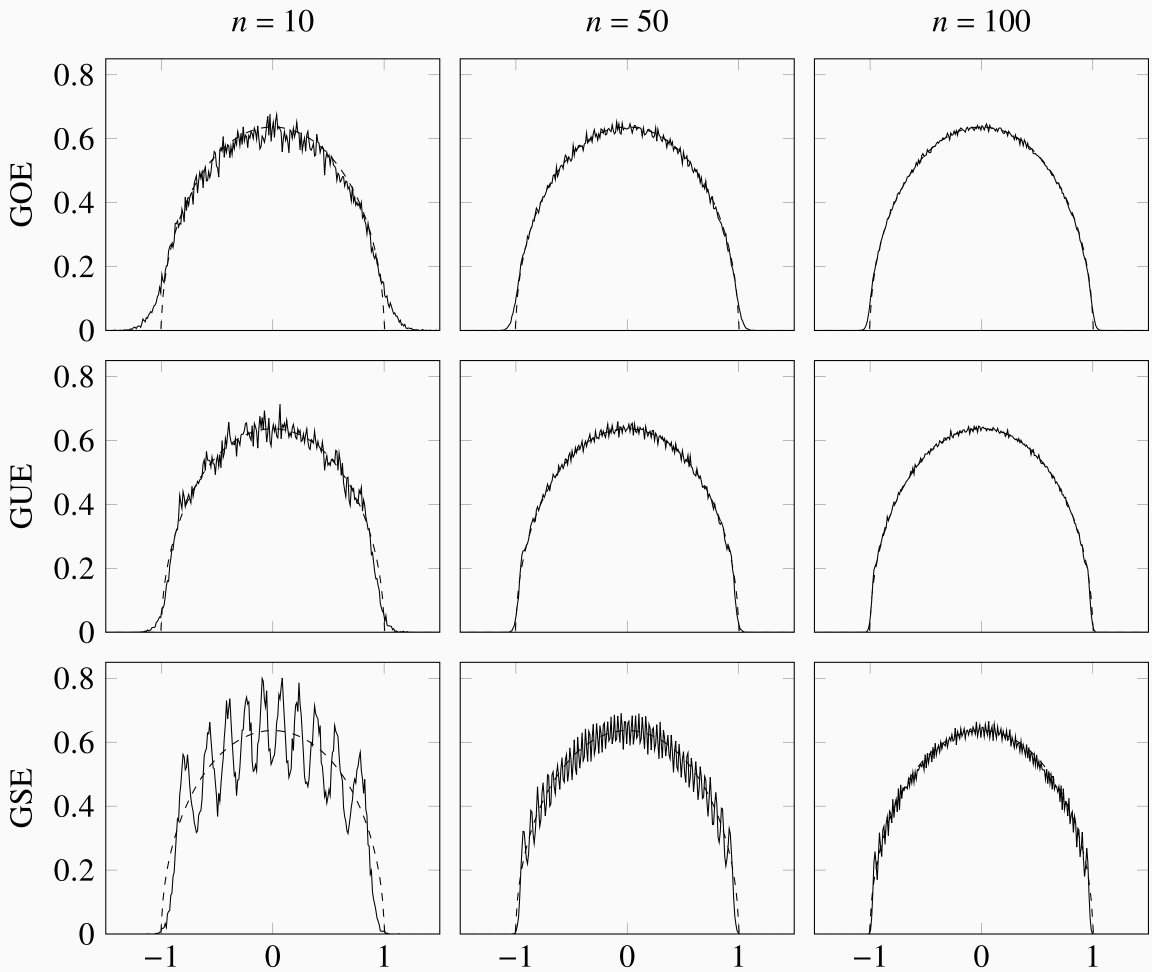

Figure 2 – The dashed line is the Semi-circle distribution, whereas in solid we display the average distribution of eigenvalues from $\hat{M}_n$. The columns correspond to $n = 10, 50, 100$ from left to right. The lines correspond to the GOE, GUE and GSE from top to bottom. We took 10000 matrix samples for each average distribution.

The second approach, of invariance, lead us to construct ensembles with invariant measure: We equip the set $\mathcal{S}$ of matrices with an $\mathcal{A}$-invariant measure $\mathcal{P}$, i.e. $\mathcal{P}(\hat{M}) = \mathcal{P}(\mathcal{A}(g, \hat{X}))$ for a group action $\mathcal{A}$. Both the *Ginibre Ensembles* introduced in Example 2.1 and the *Gaussian Ensembles* introduced in Example 2.2 are constructed considering orthogonal, unitary and symplectic conjugation invariance. Interestingly, when dealing with matrices with real eigenvalues, there is only one ensemble that is rationally invariant and has independent entries: The Gaussian ensembles. No other ensemble of Wigner matrices is also invariant (and vice-versa). Although the theory of real eigenvalues has its historical and theoretical place, our results concern complex matrices with complex eigenvalues. In the next section we discuss the details of this approach.

2.2 Invariant Ensembles

Invariant ensembles of random matrices are probability spaces invariant under some group action. However, *invariant matrix ensemble* usually refers to invariance under the conjugation mapping $\hat{X} \mapsto g\hat{X}g^*$ for g drawn from the appropriate compact group $\mathcal{G}$. *Conjugation invariance* is a very natural physical assumption and it carries its significance in mathematics; it simply means that all coordinate systems are equivalent - think of these ensembles as having probability distribution depending on their eigenvalue configurations while their eigenvectors are independent and Haar (uniformly) distributed. According to Dyson (1962), when a matrix ensemble underlying set of matrix is $\mathcal{S} \subseteq \mathbb{M}_{n,n}(\mathbb{R}), \mathbb{M}_{n,n}(\mathbb{C}), \mathbb{M}_{n,n}(\mathbb{H})$ - $n \times n$ matrices with real, complex and quaternionic entries respectively, the group $\mathcal{G}$ must be one of $O(n), U(n), Sp(2n)$ - the orthogonal, unitary and symplectic groups. Then, we say the ensemble is *Orthogonal*, *Unitary* or *Symplectic*.

Example 2.4. Consider the three ensembles from the aforementioned symmetries.
The *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \hat{M}^* = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O}\hat{M}\hat{O}^T$, $\hat{O}$ orthogonal, i.e. $\hat{O}\hat{O}^T = \hat{O}^T\hat{O} = \hat{I}$.
The *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}^*$, $\hat{U}$ unitary, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$.
The *Symplectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J}\hat{M}^T\hat{J}^T$ where $\hat{J} = \operatorname{diag}(\sigma, \cdots, \sigma)$ with $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$ and $\hat{U}\hat{J}\hat{U}^T = \hat{J}$.

For this work, we are mainly concerned with statements concerning ensembles constructed with *normal matrices* of complex entries, and therefore, with *unitary invariance*. Then, we make formal the following definition of *invariance*.

Definition 2.3. *(Invariance) Given a group $\mathcal{G}$ and a group action $\mathcal{A} : \mathcal{G} \times \mathcal{S} \to \mathcal{S}$, a random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is said to be **$\mathcal{A}$-invariant** if the probability measure* $\mathrm{d}\mathcal{P}(\hat{X}) := \mathcal{P}(\hat{X})\mathrm{d}\hat{X}$ *satisfies*

$$\mathrm{d}\mathcal{P}(\mathcal{A}(g, \hat{X})) = \mathrm{d}\mathcal{P}(\hat{X})$$

*for every $g \in \mathcal{G}$. In our particular case we say $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ to be **invariant** if the probability measure* $\mathrm{d}\mathcal{P}(\hat{X})$ *satisfies*

$$\mathrm{d}\mathcal{P}(\hat{U}\hat{X}\hat{U}^*) = \mathrm{d}\mathcal{P}(\hat{X})$$

for every $\hat{U} \in U$, the unitary group of matrices.

To construct ensembles with unitary invariance we need to take $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ such that $\mathrm{d}\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})\mathrm{d}\hat{X}$ is invariant under unitary conjugation. This is to say that we should establish that both the p.d.f. $\mathcal{P}(\hat{X})$ and the Lebesgue measure $\mathrm{d}\hat{X}$ are invariant under the aforementioned action. Invariance of the function $\mathcal{P}(\hat{X})$ is straightforward if we take

$$\mathcal{P}(\hat{X}) = \phi\left(\mathrm{Tr}\, Q(\hat{X})\right) \tag{2.2}$$

for some function ϕ in terms of the trace of a polynomial Q of the matrix. Note that, because of the trace properties, if $\hat{A} = \hat{U}\hat{B}\hat{U}^*$, then $\mathcal{P}(\hat{A}) = \mathcal{P}(\hat{B})$ since

$$\phi\left(\mathrm{Tr}\, Q(\hat{A})\right) = \phi\left(\mathrm{Tr}\, Q(\hat{U}\hat{B}\hat{U}^*)\right) = \phi\left(\mathrm{Tr}\, Q(\hat{U}^*\hat{U}\hat{B})\right) = \phi\left(\mathrm{Tr}\, Q(\hat{B})\right).$$

Therefore, if a p.d.f. is such as in (2.2), then it is invariant by our definition. Moreover, since $\hat{X}$ will be a normal $n \times n$ matrix, we know by the spectral theorem that it is similar to a diagonal matrix by unitary conjugation. Therefore, for some $\Lambda = \mathrm{diag}\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, ϕ takes the form

$$\phi\left(\mathrm{Tr}\, Q(\hat{X})\right) = \phi\left(\mathrm{Tr}\, Q(\hat{U}\hat{\Lambda}\hat{U}^*)\right) = \phi\left(\mathrm{Tr}\, Q(\hat{\Lambda})\right) = \phi\left(\sum_i^n Q(\lambda_i)\right).$$

The invariance of the Lebesgue measure is more involved and depends on whether or not the associated matrix set $\mathcal{S}$ consists of purely self-adjoint matrices. The extension for complex matrices was obtained by Ginibre (1965) exploring the now called *Ginibre ensembles*. Ginibre extended the classical argument used for Hermitian matrices to Ginibre matrices $\hat{G}$ using the diagonalization formula $\hat{G} = \hat{P}\hat{\Lambda}\hat{P}^{-1}$. Note that this is valid with probability one since the subspace of matrices with simple spectra (unique eigenvalues) is dense on the space of complex matrices and has full Lebesgue measure.

Proposition 2.1. *The set of complex matrices with simple spectra is open and dense in $\mathcal{M}_n$ (n-dimensional complex matrices) and has full measure, that is, generally, complex matrices have simple eigenvalues.*

Proof. Let the matrix $\hat{M}$ have the decomposition

$$\hat{M} = \hat{P}\hat{\Lambda}\hat{P}^{-1},$$

with $\hat{\Lambda}$ diagonal with eigenvalues $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ and $\hat{P}$ a matrix of eigenvalues. To prove denseness of matrices with simple eigenvalues we note that there exists real variables $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$, with $\sum_{i=1}^n |\epsilon_i| \to 0$, such that

$$\hat{M}_\epsilon = \hat{P}\begin{bmatrix} \lambda_1 + \epsilon_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n + \epsilon_n \end{bmatrix}\hat{P}^* = \hat{M}_\epsilon^*$$

has a simple spectrum. Because $\hat{M}_\epsilon \to \hat{M}$, matrices with simple spectra are dense in $\mathcal{M}_n$, the space of n-dimensional normal matrices. Furthermore, to show that the set of matrices with simple eigenvalues has full measure we shall prove that its complement is union of lower dimensional submanifold. First, consider the map

$$\begin{aligned} f_i \colon \mathcal{M}_n &\to \mathbb{C} \\ \{m_{i,j}\} &\mapsto a_i\left(\{m_{i,j}\}\right) \end{aligned}$$

where a_i is the i-th coefficient for the characteristic polynomial of $\hat{M}$ function of its entries. However, knowing the roots of such polynomial to be a certain set $\{\lambda_i\}_{i=1}^n$, possibly repeated, one can use Vieta's Formulas to write the coefficient as the fundamental symmetric polynomial on the roots, i.e.

$$a_i\left(\{m_{i,j}\}\right) = \varsigma_i(\lambda_1, \lambda_2, \dots, \lambda_i) := \sum_{1\le k_1 \le \dots \le k_i \le n} \lambda_{k_1}\lambda_{k_2}\dots\lambda_{k_i} \tag{2.3}$$

for $i = 0, \dots, n-1$. Introduce now the discriminant defined by

$$\begin{aligned} D \colon \mathbb{C}^n &\to \mathbb{R} \\ \{\lambda_1, \dots, \lambda_n\} &\mapsto \prod_{i<j} |\lambda_i - \lambda_j|^2. \end{aligned}$$

We note that this is a symmetric function. As such, by the fundamental theorem of symmetric function we can write it as

$$D(\Lambda) = P\left(\varsigma_1(\Lambda), \dots, \varsigma_n(\Lambda)\right)$$

where $\{\lambda_1, \lambda_2, \dots, \lambda_n\} = \Lambda$. Because of (2.3), we can also see D as a polynomial function of the entries of the matrix, i.e

$$D(\hat{M}) = P\left(a_0\left(\{m_{i,j}\}\right), \dots, a_{n-1}\left(\{m_{i,j}\}\right)\right).$$

This is a multivariate polynomial on the entries $m_{i,j}$ of the matrix. We now think of this as a polynomial of one of the variables, such as $m_{1,1}$, and factor out $l_{i,j}$ factors of the type $(m_{i,j} - f_{i,j}^{(1)}(\{m_{l,k}\}_{k,l\neq i,j}))$. Note that $l_{i,j}$ is finite and may be zero for some of the entries. Using this, we write

$$D(\hat{M}) = \prod_{i,j}\prod_{k=1}^{l_{i,j}} (m_{i,j} - f_{i,j}^{(k)}(\{m_{l_1,l_2}\}_{l_1,l_2\neq i,j})) := \prod_{i,j}\prod_{k=1}^{l_{i,j}} p_{i,j}^{(k)}$$

Note also that, denoting the set of zeros for a function f as $Z(f)$, we must have by definition

$$Z(D) = \bigcup_{i,j,k} Z\left(p_{i,j}^{(k)}\right)$$

and more importantly, the set of matrices with simple eigenvalues is precisely $\mathcal{M} \setminus Z(D)$. Note that each of the polynomials $p_{i,j}^{(k)}$ define a submersion between the manifold of these matrices and the real line since

$$\frac{\partial p^{(k)} i,j}{\partial m_{i,j}} \neq 0.$$

Therefore, by the local form of the submersion theorem $Z\left(p_{i,j}^{(k)}\right)$ is a manifold with dimension n^2-1 and must have null measure. The same holds for the union of all these function and consequently for $Z(D)$. □

For the ensembles we take interest in this work, we only need to consider the result regarding the measure $\mathrm{d}\hat{X}$ on the set of normal matrices with complex entries $\mathcal{N}(\mathbb{C})$. We now state this result, which we call the *Weyl's integration formula*.

Theorem 2.2. *(Weyl's integration formula - Unitary case) The measure* $\mathrm{d}\hat{X}$ *decomposes almost everywhere in* $\mathcal{N}(\mathbb{C})$ *as*

$$\mathrm{d}\hat{X} = \mathrm{d}\hat{U} \prod_{1\leq i<j\leq n} |z_i - z_j|^2 \prod_{i=1}^{n} \mathrm{d}^2 z_i,$$

where $\{z_i\}_{i=1}n$ *is the spectrum of* $\hat{X}$, $\hat{M} = \hat{U}\hat{\Lambda}\hat{U}^*$ *for* $\hat{\Lambda} = \mathrm{diag}\{z_i\}_{i=1}^n$, $\hat{U} \in U(n)/U(1)^n$ *and the measure*

$$\mathrm{d}\hat{U} = 2^{\frac{n(n-1)}{4}} \prod_{i<j} \mathrm{d}^2 u_{i,j}, \quad \mathrm{d}^2 u_{i,j} = (\hat{U}^*)_{i,k} \mathrm{d}^2 \hat{U}_{k,j}$$

on $U(n)/U(1)^n$ *is induced by the measure on* $U(n)$ *and is a Haar measure.*

With Theorem 2.2, we can directly check that $\mathrm{d}\hat{X}$ is invariant by its decomposition. Naturally, $\prod_{1\leq i<j\leq n} |z_i - z_j|^2 \prod_{i=1}^n \mathrm{d}^2 z_i$ is invariant under unitary conjugation as it is a function of the spectrum of $\hat{X}$. Moreover, $\mathrm{d}U$ is a Haar measure, i.e. has exactly the properties of invariance by left and right unitary transformations. While eigenvectors may be of interest, eigenvalue statistics have been noticeably more explored; justified by the uniform distribution on its eigenvectors. This gives a reason for the larger interest in eigenvalue statistics. Before advancing we discuss some objects that are fundamental when studying RMT; namely, we introduce the joint probability density function, the correlation function and the univariate density.

2.3 Eigenvalue Statistics

We are mainly interested in studying eigenvalue configurations. This is especially true because we are dealing with ensembles which the p.d.f. is uniform in the eigenvectors. Interestingly, setting a joint probability density function (j.p.d.f.) on a configuration of points is the building block of a commonly know and studied structure called *point process*. Suppose there are n eigenvalues at random locations in some complete separable metric space Λ - $\mathbb{C}$ in our context - denoted by the configuration $\chi_n = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$. We define a quantity $\xi(A)$, similar to (2.1), that measures the amount of eigenvalues ($\#\{\chi_n \cap A\}$) for any sub-region $A \subset \mathbb{C}$,

$$\xi(A) = \sum_{k=1}^{n} \mathbb{1}_A(\lambda_k).$$

As before, this defines a *random counting measure* in the usual sense. We say that the counting measure ξ is simple if $\xi(\{x\}) \leq 1$ for all $x \in \mathbb{C}$. Denote $\mathcal{E}$ denote the class of Borel sets and $\mathcal{N}(\mathbb{C})$ the space of all counting measures on $\mathbb{C}$ - also the space of configurations in a sense.

Definition 2.4. *(Point Processes on the complex plane) A* ***point process*** $\mathbb{P}$ *on* $\mathbb{C}$ *is a probability measure on the space of all counting measures* $\mathcal{N}(\mathbb{C})$*. We say* $\mathbb{P}$ *to be a n-point process if* $\mathbb{P}(\xi(\mathbb{C}) = n) = 1$*. We also say that the point process is simple if* $\mathbb{P}(\xi$ *is simple*$) = 1$.

We argue that the ensemble p.d.f. of invariant ensembles will induce a joint probability density function on the space of configurations of eigenvalues, i.e. will induce a simple point process (it is simple because of Proposition 2.1). In fact, note that the mapping $\mathcal{E} \ni A \mapsto \mathbb{E}[\#(\chi \cap A)]$, that assigns to the Borel set $A \in \mathbb{C}$ the expected value of number of points in itself, under the configuration χ, is a measure on $\mathbb{C}$. Although it is not true for general point processes, in the models we are interested in working with, this measure can be defined as the integral of a function we call 1*-point correlation function*. This function measures the probability of finding a particle at a given infinitesimal.

Definition 2.5. *The* ***1-point correlation function*** p_1 *represents the infinitesimal probability of a point of the process being at a point z, i.e.* $\mathbb{P}(\xi(\mathrm{d}^2z) = 1) = \mathrm{p}_1(z)\mathrm{d}^2z$*. It is such that the* ***mean measure****, or* ***first moment****, of a point process on* $\mathbb{C}$*, defined by*

$$p(A) = \mathbb{E}[\xi(A)] := \mathbb{E}[\#(\chi \cap A)], \quad A \in \mathcal{E}.$$

is given by its integral over A

$$p(A) = \int_A \mathrm{p}_1(z)\mathrm{d}A$$

where $\mathrm{d}A = \mathrm{d}^2z$ *is the area measure.*

In RMT, the usual name for the 1-point correlation function p_1, when normalized, is *univariate eigenvalue density*, for which we use the symbol $\mu = \mathrm{p}_1/n$ henceforth. Moreover, we can also may be interested in the infinitesimal probability that some k points of the process are in a certain configuration $\chi_k = \{z_1, z_2, \ldots, z_k\}$ for k in between 2 and n. For this we define more generally all the *k-th point correlation functions* and *k-th moments*.

Definition 2.6. *The* ***k-point correlation function*** p_k *represents the infinitesimal probability of k points of the process being at a k-configuration χ_k. It is defined such that the* ***k-th moment*** *of a point process on* $\mathbb{C}$*, defined by*

$$p_k(A_1 \times A_2 \times \cdots \times A_k) := \mathbb{E}\left[\xi(A_1)\xi(A_2)\cdots\xi(A_k)\right], \quad A_1, A_2, \ldots, A_k \in \mathcal{E},$$

where $A_j \cap A_i = \varnothing$ for $i \neq j$ is given by the integral

$$p_k(A_1 \times A_2 \times \cdots \times A_k) = \mathbb{E}\left[\prod_{j=1}^{k}(\#\chi \cap A_j)\right] = \int_{A_1} \cdots \int_{A_k} \mathrm{p}_k(z_1, \cdots, z_k)\mathrm{d}A_1 \cdots \mathrm{d}A_k.$$

where $\mathrm{d}A_i = \mathrm{d}^2 z_i$ is the area measure.

For simple point processes on $\mathbb{C}$ there is a direct way to understand correlation functions as mentioned, $\mathrm{p}_k(z_1, z_2, \ldots, z_k)$ is exactly the probability of finding particles at positions $x_1, x_2, \ldots, x_k$. This means that $\mathrm{p}_1(z)$ is the density of particles at z and a similar argument can be made for $\mathrm{p}_k(z_1, z_2, \ldots, z_k)$. Now, if you have some $p_n(z_1, z_2, \ldots, z_n)$ a j.p.d.f. on $\mathbb{C}^n$, invariant under permutation of coordinates, then we can build a n-point process on $\mathbb{C}$ with correlation functions

$$\mathrm{p}_k(z_1, z_2, \ldots, z_k) = \frac{n!}{(n-k)!}\int_{\mathbb{C}^{n-k}} p_n(z_1, z_2, \ldots, z_n)\mathrm{d}A_{k+1} \cdots \mathrm{d}A_n. \tag{2.4}$$

Moreover,

$$p_n = \frac{1}{n!}\mathrm{p}_n.$$

Example 2.5. Consider the induced point process on $\Lambda = \mathbb{R}$ given by the Gaussian orthogonal ensemble, introduced in Example 2.2. We have the ensemble $\mathcal{E} = (\mathcal{H}_n, \mathcal{P}^{(G)})$ with the space of all $n \times n$ real Hermitian matrices and $\mathcal{P}^{(G)}$ the corresponding Gaussian measure. We know that for $\hat{M} \in \mathcal{H}_n$,

$$\hat{M} \mapsto \sum_{j=1}^{n} \mathbb{1}(\lambda_j(\hat{M}))$$

maps $\mathcal{P}^{(G)}$ to a finite point process $\mathbb{P}$ on $\mathbb{R}$. If $\mathrm{d}\hat{M}$ is the Lebesgue measure on $\mathcal{H}_n$,

$$\mathrm{d}\mathcal{P}(\hat{M}) = \frac{1}{\mathcal{Z}_n^{(G)}}\mathrm{e}^{-n\operatorname{Tr}\hat{M}^2}\mathrm{d}\hat{M} = \frac{1}{\mathcal{Z}_n^{(G)}}\mathrm{e}^{-n\operatorname{Tr}\hat{\Lambda}^2}\mathrm{d}\hat{M} = \frac{1}{\mathcal{Z}_n^{(G)}}\mathrm{e}^{-n\sum_i^n \lambda_i^2}\mathrm{d}\hat{M}$$

where $\{\lambda_1(\hat{M}), \cdots, \lambda_n(\hat{M})\}$ denotes the set of eigenvalues of $\hat{M} \in \mathcal{H}_n$. It can be shown that for any symmetric, continuous function f on $\mathbb{R}^n$ with compact support (see Theorem 2.2) it holds that $\mathrm{d}M = \prod_{1\le i<j\le n}^{n}|\lambda_i - \lambda_j|\mathrm{d}\Lambda\mathrm{d}\hat{U}$. Then the j.p.d.f. on the eigenvalues is

$$p_n(\lambda_1, \lambda_2, \ldots, \lambda_k) = \frac{1}{\mathcal{Z}_n^{(G)}}\prod_{1\le i<j\le n}^{n}|\lambda_i - \lambda_j|\prod_{j=1}^{n}\mathrm{e}^{-n\lambda_j^2}$$

Therefore the point process $\mathbb{P}$ on $\mathbb{R}$ defined by the Gaussian orthogonal p.d.f. has correlation functions

$$\mathrm{p}_k(x_1, x_2, \ldots, x_k) = \frac{n!}{\mathcal{Z}_n^{(G)}(n-k)!}\int_{\mathbb{R}^{n-k}}\prod_{1\le i<j\le n}^{n}|x_i - x_j|\prod_{j=1}^{n}\mathrm{e}^{-nx_j^2}\prod_{i=k+1}^{n}\mathrm{d}x_i,$$

analogous to the expression given by (2.4).

2.4 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those with p.d.f.s $\mathcal{P}(\hat{X})$ such that one has the decomposition

$$\mathcal{P}(\hat{X}) = \frac{1}{\mathcal{Z}_n^w}|\Delta_n(\lambda)|^2 \prod_{i=1}^{n} w(\lambda_i)$$

where $\{\lambda_i\}$ are the indistinguishable eigenvalues of $\hat{X}$, $w(\lambda)$ is some continuous weight function that is real valued and decays fast enough as $|\lambda| \to \infty$ and $|\Delta_n(\lambda)|$ is the Vandermonde determinant. The second condition on the weight function guarantees that normalization constant for the j.p.d.f. considered is well-defined. We are interested in the probability distribution of the eigenvalues $\lambda_1, \cdots, \lambda_n$ of $\hat{X}$. By the spectral theorem, for any normal matrix we have $\hat{U}$ an unitary matrix such that $\hat{U}^*\hat{X}\hat{U} = \hat{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

To arrive at the expression for the eigenvalue j.p.d.f. we integrate the eigenvector dependent terms so that we are left with only the part depending on the eigenvalues. This integral on $\mathcal{P}$ can be evaluated using Theorem 2.2 and the fact that $\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{\Lambda})$ for $\hat{X} = \hat{U}\hat{\Lambda}\hat{U}^*$. Then, using Theorem 2.2 we show that the integral on the quotient space with respect to $\mathrm{d}\mathcal{P}$ is given by

$$\begin{aligned}
\mathrm{d}p_n(\{\lambda_i\}_{i=1}^n) = \int_{U(n)/U(1)^n} \mathcal{P}(\hat{X})\mathrm{d}\hat{X} &= \frac{1}{\mathcal{Z}_n^{\mathcal{P}(\Lambda)}}\mathcal{P}(\hat{\Lambda}) \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=1}^{n} \mathrm{d}^2\lambda_i \\
&= \frac{1}{\mathcal{Z}_n^w} \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=1}^{n} w(\lambda_i) \prod_{i=1}^{n} \mathrm{d}^2\lambda_i.
\end{aligned}$$

where we took $\mathcal{P}(\Lambda) = \prod_{i=1}^{n} w(\lambda_i)$ and

$$\mathcal{Z}_n^w = \int_{\mathbb{C}^n} \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=1}^{n} \prod_{i=1}^{n} w(\lambda_i)\mathrm{d}^2\lambda_i.$$

Example 2.6. (Gaussian Measure) Let us make it explicit the weight function defining the Gaussian Unitary Ensembles introduced in Example 2.2. If U_n is the unitary group, we know by the previous calculation that

$$\begin{aligned}
\int_{U_n} \mathcal{P}(\hat{M})\mathrm{d}\hat{M} &= \int_{\mathbb{V}_n} \mathcal{P}(\hat{\Lambda}, \hat{U}) \prod_{i<j} |\lambda_j - \lambda_i|^2 \mathrm{d}\hat{U}\mathrm{d}\hat{\Lambda}, \\
&= \frac{1}{\mathrm{Vol}(U_n^{-1}} \mathcal{P}(\hat{\Lambda}) \prod_{i<j} |\lambda_j - \lambda_i|^2 \prod_{i=1}^{n} \mathrm{d}^2\lambda_i
\end{aligned}$$

where $\hat{M} = U\Lambda U^*$. We also have seen that

$$\mathcal{P}(\hat{M}) = \exp\left\{-n\,\mathrm{Tr}\left(\hat{H}^2\right)\right\} = \exp\left\{-n\,\mathrm{Tr}\left(\hat{\Lambda}^2\right)\right\} = \prod_{i=1}^{n} \mathrm{e}^{-n\lambda_i^2} = \mathcal{P}(\hat{\Lambda})$$

which says that $w(x) = \mathrm{e}^{-nx^2}$ and lead us to write the j.p.d.f. as in Example 2.5

Definition 2.7. *Fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative, real-valued function such that $w(\lambda) = o(|\lambda|^{-2n+1})$. Then, the j.p.d.f.*

$$\mathrm{p}_n(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{1}{\mathcal{Z}_n^w} |\Delta_n(\lambda)|^2 \prod_{i=1}^{n} w(\lambda_i) \tag{2.5}$$

where $|\Delta_n(\lambda)|$ is the Vandermonde determinant, describes the n-sized ***orthogonal polynomial ensemble*** *(OPE).*

If we stare at (2.5) strongly and for long, the terms become less mysterious. One may notice that this expression is simply the normalized product of some weight function $w(\lambda)$ and the Jacobian of the transformation $\hat{M} \mapsto \hat{U}\hat{\Lambda}\hat{U}^*$ given by the squared Vandermonde determinant. More than that, there is an observation to be made here in terms of the behavior of the eigenvalues. Notice that the Jacobian takes arbitrarily small values if there are equally arbitrarily close eigenvalues in the set. At first instance this means that our set is most likely simple, i.e. without repeated eigenvalues. Moreover it also means that the eigenvalues show some repulsion with each-other. Why then one might not expect the eigenvalues to scatter in the line? This is the role of the weight function. More than characterizing the ensemble, in the formula it has the role of ensuring integrability, i.e. holding the eigenvalues in a limited set. It must, therefore, decay sufficiently fast. If this feels like some physical repelling gas system, fear not, this interpretation is something we make use of and will be expanded in section 2.5.

Example 2.7. A random matrix ensemble is said to be a unitary *Classical Matrix Ensemble* if its eigenvalue j.p.d.f. is of the form

$$\mathrm{p}_n(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{1}{\mathcal{Z}_n^w} \prod_{i=1}^{n} w(\lambda_i) |\Delta_n(\lambda)|^2,$$

where $\Delta_n(\lambda)$ is the Vandermonde determinant and w, the weight function, has among others, one of the following forms

$$w(\lambda) = \begin{cases} \mathrm{e}^{-n\lambda^2} & \text{Gaussian,} \\ n^{\frac{a}{2}} \lambda^a \mathrm{e}^{-n\lambda} \mathbb{1}_{\{\lambda > 0\}} & \text{Laguerre,} \\ n^{\frac{a}{2}} \lambda^a (1 - \sqrt{n}\lambda)^b \mathbb{1}_{\{0 < \lambda < n^{-1/2}\}} & \text{Jacobi.} \end{cases}$$

where $\mathbb{1}$ is the indicator function and $a, b > -1$. All of the classical ensembles defined as is are invariant under unitary conjugation. Moreover, you could define these ensembles with the other two conjugation invariance changing the Jacobian ot get orthogonal symplectic conjugation invariance. The names Hermite, Laguerre and Jacobi also refer to a class of polynomials, the reason for this and the ensemble's name will become clearer in section 2.6.

2.5 Coulomb Gases

We start by describing a familiar distribution for physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of n particles as a function of the system state's *energy* ($\mathcal{H}$) and the *temperature* ($1/(\beta n^2)$). Physically, this is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy. This distribution is common on thermodynamics by this very reason. Let's introduce a system that has states partitioned accordingly to such distribution. Suppose a set of n particles at positions $x_1, x_2, \ldots, x_n \in \Sigma$, a *conductor* of dimension d, or in some configuration $\chi_n = (x_1, x_2, \ldots, x_n)$, embedded $\mathbb{R}^{dn}$.

Definition 2.8. *If* dp_n *is a probability measure on* $(\mathbb{R}^d)^n$*, we name it* ***Coulomb gas*** *if it has the form of a Boltzmann-Gibbs probability measure, namely if*

$$\mathrm{dp}_n(\chi_n) = \frac{\mathrm{e}^{-\beta n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_{n,\beta}^{Q}} \prod_{i=1}^{n} \mathrm{d}x_i, \tag{2.6}$$

where

$$\mathcal{H}(\chi_n) = \frac{1}{2n}\sum_{i=1}^{n} Q(x_i) + \frac{1}{n^2}\sum_{i<j}^{n} \mathfrak{g}(x_i - x_j)$$

is usually called the ***Hamiltonian****, or more concretely, the* ***configurational energy*** *of the system. The function* $\mathfrak{g}\colon \Sigma \to (-\infty, \infty]$ *is the Coulomb Interacting Kernel, solution to the Poisson equation and given by* $\mathfrak{g}(\vec{x}) = -\ln(|\vec{x}|)$ *for* $d = 1, 2$*. The function* $Q\colon \mathbb{R}^d \cong \Sigma \to \mathbb{R}$ *is the external potential and* $\mathcal{Z}_{n,\beta}^{Q}$ *is the* ***partition function****, or normalization function, given by*

$$\mathcal{Z}_{n,\beta}^{Q} = \int_{\mathbb{R}^{dn}} \prod_{i=1}^{n} \mathrm{d}x_i \, \mathrm{e}^{-\beta n^2 \mathcal{H}(\chi_n)}.$$

The measure dp_n models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q. We wish dp_n to be a probability measure, then one must ensure that Q is such that the normalizing constant $\mathcal{Z}_{n,\beta}^{Q}$ is finite for all n. This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. Recall the expression for the polynomial ensembles in Definition 2.7. As we are dealing with point processes on the plane, we take the conductor to be two dimensional ($d = 2$), $\beta = 2$ and rearrange the terms in (2.6) to get

$$\mathrm{dp}_n(x_1, x_2, \ldots, x_n) = \frac{1}{\mathcal{Z}_n^{Q}} \prod_{i<j}^{n} |x_i - x_j|^2 \prod_{j=i}^{n} \mathrm{e}^{-nQ(x_i)} \, \mathrm{d}x_i.$$

The similarity to Definition 2.7 indicates that eigenvalues from invariant ensembles behave as a Coulomb gas. This opens the way for interpretations on the distribution of eigenvalues but also to computations using simulation methods for particle dynamics. This also motives why most often the weight functions used for invariant ensembles will have the form, for some polynomial Q,

$$w(x) = \mathrm{e}^{-nQ(x)}. \tag{2.7}$$

This directly relates to the weight function in Definition 2.7 and gives a way to trap particles in a limited - set giving lesser probabilities for configurations with more diffused particles.

Example 2.8. As explored in Chafaï and Ferré (2018), a few classical algorithms of simulation are very successful in replicating the measures for Coulomb gas systems. Equivalently, if one is interested in sampling eigenvalues for our invariant ensembles, one can sample from its configuration equilibrium $\mathrm{p}_Q(\lambda)$ with respect to some potential Q using simulation of the particles. Consider for example the non-trivial case explored in Balogh, Grava and Merzi (2016), where, taken $d = 2$ and $n = 2$ we will have $W(x) = g(x) = \log|x|$ and impose the potential

$$Q(z) = |z|^{2a} - \mathrm{Re}\, tz^a.$$

This potential has disjoint intervals for some combinations of the parameters t and a - it presents a transition in $t_c \approx \sqrt{(1/a)}$. Analytically, it was just recently solved. However we can easily sample from this measure using particle simulations, as shown below. Data was taken from Pimenta (2024).

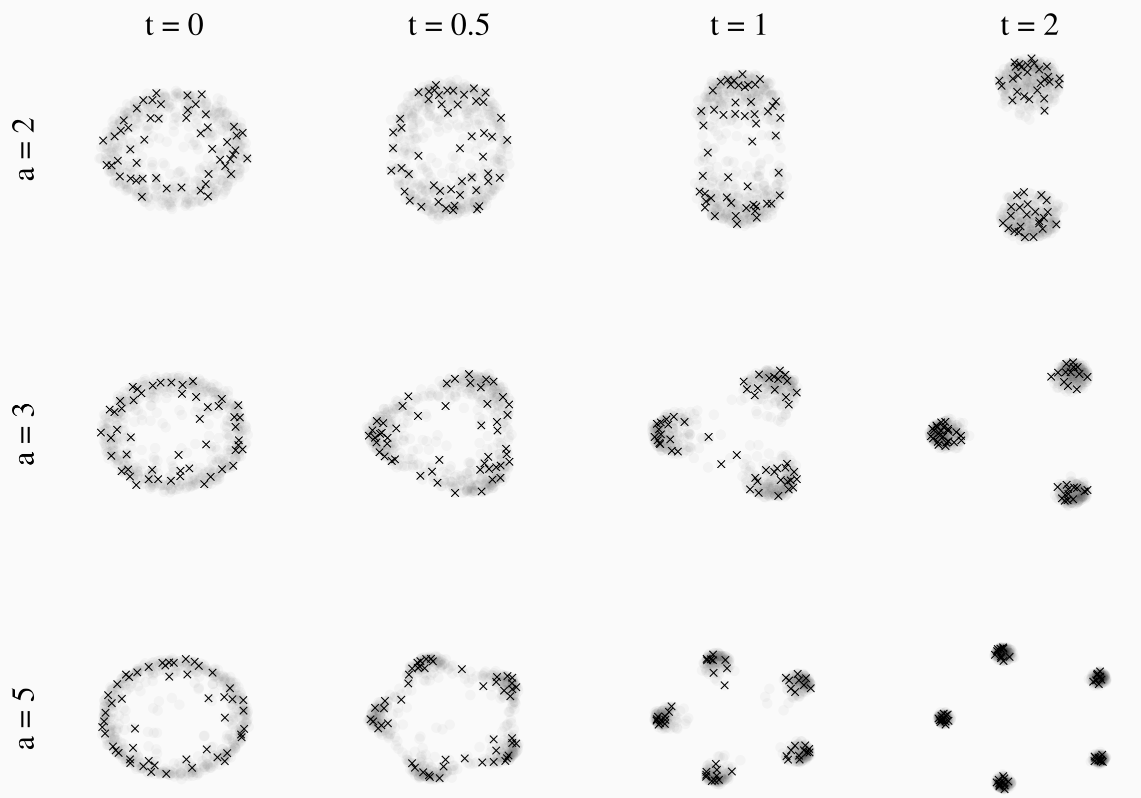

Figure 5 – Shown in the shadow is the mean configurations - in some an approximation of the limit distribution of points, and in the scatter is a random $n = 50$ sample from the distribution. We show configurations for combinations of the parameters $t = 0, 0.5, 1, 2$ from left ro right and $a = 2, 3, 5$ from top to bottom.

2.6 Determinantal Structure

We stated that random eigenvalues can be described as point processes. However, when dealing with invariant ensembles there is more structure: There seems to be repulsion between eigenvalues. Repulsive random processes are, in general, hard to study. They are usually difficult to, for example, compute marginals and conditional probabilities, making their theoretical analysis a major challenge. However, there is a class of processes that is an exception to this rule: *Determinantal point processes* - or Fermionic point processes, in the physics literature. These are random processes with the property of being both repulsive and tractable. (TREMBLAY, 2025) This property comes from the fact that their correlation functions can be expressed with respect to a determinant. We now make this precise.

Definition 2.9. *(Determinantal point processes on the plane) Consider a point process $\mathbb{P}$ on $\mathbb{C}$, with joint probability density function measure p_n, all of whose correlation functions p_k exist. If there is a function $K : \mathbb{C}\times\mathbb{C} \to \mathbb{C}$ such that*

$$\mathrm{p}_k(z_1, z_2, \dots, z_k) = \det(K(z_i, z_j))_{i,j=1}^{k}$$

*for all $z_1, z_2, \dots, z_k \in \mathbb{C},\ k \geq 1$, then we say that $\mathbb{P}$ is a **determinantal point process** on $\mathbb{C}$. We call K the **correlation kernel** of the process.*

This definition can be extended for a general complete separable metric space but its full extent is outside of our scope. To more directly state results and focus on our main ensembles, we focus on weight functions as is (2.7). We remember that the correlations functions for the polynomial ensemble are given by the expression

$$\mathrm{p}_k(\lambda_1, \lambda_2, \dots, \lambda_k) = \frac{n!}{\mathcal{Z}_n^Q (n-k)!} \int_{\mathbb{C}^{n-k}} \prod_{i=1}^{n} \mathrm{e}^{-nQ(\lambda_i)} \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=k+1}^{n} \mathrm{d}^2\lambda_i.$$

To show that we can find a function $K : \mathbb{C}\times\mathbb{C} \to \mathbb{C}$ such that $\mathrm{p}_k(x_1, x_2, \dots, x_k) \propto \det(K(x_i, x_j))_{i,j=1}^{k}$, we define *orthogonal polynomials* and prove a proposition regarding the determinant $|\Delta_n(\lambda)|$.

Definition 2.10. *With Q a suitable function, define the following inner products on polynomial space:*

$$\langle f, g\rangle_n = \int_{\mathbb{C}} \overline{f(z)} g(z)\, \mathrm{e}^{-nQ(z)}\, \mathrm{d}^2 z.$$

Let $\{q_{j,n}\}_{j=0}^{\infty}$ be a family of monic polynomials such that each $q_j(\lambda)$ is of degree j. Impose also that exists finite positive constants $\{r_{j,n}\}_{j=0}^{\infty}$ s.t.

$$\langle q_{j,n}, q_{k,n}\rangle_n = r_{j,n} \mathbb{1}_{\{j=k\}}$$

*for any $j, k > 0$. We say that the family $\{q_{j,n}(z)\}_{j=0}^{\infty}$ **is orthogonal with respect to the weight function** $w(z) = \mathrm{e}^{-nQ(z)}$.*

Note 2.1. (Expected characteristic polynomial for Gaussian Unitary Ensemble (GUE)) To calculate eigenvalues of some given matrix, one need to determine the *characteristic polynomial*. However, dealing with a random matrix $\hat{M}$, one of the quantities one might want to take on is the *average* or *expected characteristic polynomial*

$$P_n(x) = \mathbb{E}[\det\left(x\hat{I}_n - \hat{M}\right)], \quad n \in \mathbb{N}.$$

The characteristic polynomial has roots that coincide with the eigenvalues of the matrix. Therefore, morally, roots of an averaged characteristic polynomial can be thought of as the typical set of eigenvalues for the ensemble of $\hat{M}$. Taking the asymptotic on n - consequently the limit for the degree of the expected characteristic polynomial - the distribution of roots is expected to coincide with the distribution of eigenvalues.
Despite of this not being true for ensembles with non-real roots, for random Hermitian matrices distributed accordingly to some *weight function* w, the average characteristic polynomial is simply the monic orthogonal polynomial $P_n^w(x)$ of degree n with respect to $w(x)$ - as shown in Edelman (1989). The asymptotic on the appropriate orthogonal polynomials reveals information on the distribution of eigenvalues. The idea of orthogonal polynomials will come back as a way to express eigenvalue correlation functions.
We will indicate in Example 2.9 that *Hermite Polynomials* are the orthogonal polynomials with respect to the weight function $w(x) = \mathrm{e}^{-nx^2}$, which is the defining weight function of the Gaussian Ensemble. That means that for $\hat{M}$ a hermitian matrix of dimension n with probability density function given by

$$\frac{1}{\mathcal{Z}_n^{(G)}} \mathrm{e}^{-n \operatorname{Tr} \hat{M}^2} \mathrm{d}M,$$

one can show numerically that the concentration of roots of the appropriately normalized Hermite polynomial and the eigenvalues of $\hat{M}$ coincide in distribution; the roots represent the typical set of eigenvalues.

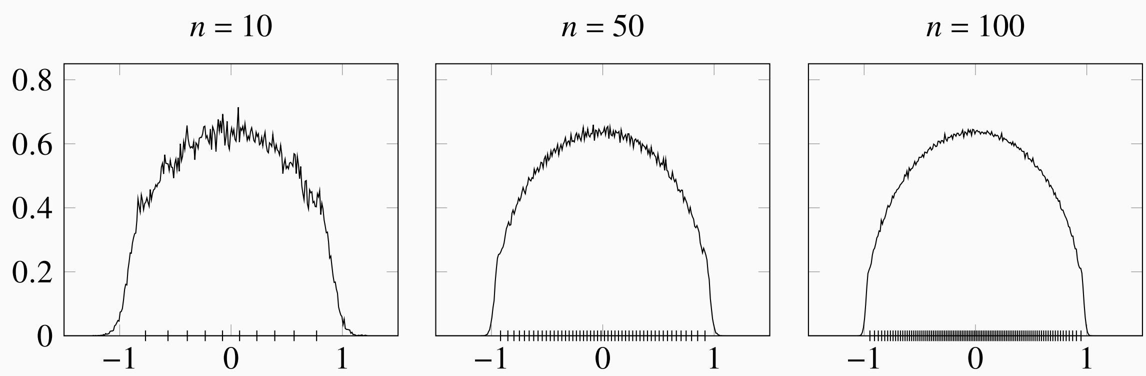

Figure 6 – For all graphs, the line is the average distribution of eigenvalues for the GUE. The columns correspond to the polynomial order (and matrix size) $n = 10, 50, 100$ from left to right. The dashes in the axis correspond to the roots of the n-th Hermite polynomial.

Now, we prove the following proposition such that we can use these orthogonal polynomials to construct the orthogonal polynomial Kernel in our correlation functions.

Proposition 2.2. *The product of the differences of the set $\{\lambda_i\}_{i=1}^n$ can be evaluated as a determinant known as the Vandermonde determinant. Moreover, it can be rewritten as the determinant of $\left(q_{j-1}(\lambda_i)\right)_{j,i=1}^n$. That is,*

$$\prod_{i<j}^{n} |\lambda_i - \lambda_j| = \det\left(q_{j-1}(\lambda_i)\right)_{n\times n}$$

where q_i are orthogonal polynomials with respect to some real valued weight function w.

Proof. Consider a matrix $\hat{V}$ given by

$$\hat{V} = \begin{bmatrix} 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \dots & \alpha_2^{n-1} \\ 1 & \alpha_3 & \alpha_3^2 & \dots & \alpha_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \dots & \alpha_n^{n-1} \end{bmatrix}.$$

We can use the determinant's multilinearity to subtract from each row r_i the product $\alpha_1 r_{i-1}$ starting from $i = n$ and ending at $i = 2$. Then subtract from every line the first line such that the first columns is exclusively zeros. Then, we factor out the term $\alpha_i - \alpha_1$ from each line, i.e.

$$|\hat{V}| = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & \alpha_2-\alpha_1 & \alpha_2(\alpha_2-\alpha_1) & \dots & \alpha_2^{n-2}(\alpha_2-\alpha_1) \\ 0 & \alpha_3-\alpha_1 & \alpha_3(\alpha_3-\alpha_1) & \dots & \alpha_3^{n-2}(\alpha_3-\alpha_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \alpha_n-\alpha_1 & \alpha_n(\alpha_n-\alpha_1) & \dots & \alpha_n^{n-2}(\alpha_n-\alpha_1) \end{vmatrix} = \prod_{i=2}^{n} |\alpha_i - \alpha_1| \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 0 & 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}.$$

Using *Laplace Theorem* for the determinant we know that there is only one co-factor that matters. Which give us

$$|\hat{V}| = \prod_{i=2}^{n} |\alpha_i - \alpha_1| \begin{vmatrix} 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}.$$

Note that the matrix is very similar to its initial form again. Repeating this procedure many times we get the expression we started with, that is,

$$|\hat{V}| = \prod_{i<j}^{n} |\alpha_j - \alpha_i|$$

Now, it is only a matter of showing that

$$|\alpha_i^j| = \left|q_{j-1}(\alpha_i)\right|$$

but this is direct since it only takes fundamental operation, i.e. by multilinearity, the operations $\alpha_i^{j-1} \mapsto q_{j-1}(\alpha_i)$ do not change the determinant. Therefore for any polynomial base $\{q_j\}_0^{n-1} := \{\sum_{i=0}^{j-1} a_{i,j}\alpha_j^i\}$, we can, by column operations of the type $R_j \mapsto \sum_{i=0}^{j-1} a_{i,j}R_j$, rewrite

$$|\hat{V}| = \begin{vmatrix} 1 & a_{1,2}\alpha_1 & \sum_{i=0}^{2-1} a_{1,j}\alpha_1^i & \dots & \sum_{i=0}^{n-1} a_{1,j}\alpha_1^i \\ 1 & a_{2,2}\alpha_2 & \sum_{i=0}^{2-1} a_{2,j}\alpha_2^i & \dots & \sum_{i=0}^{n-1} a_{2,j}\alpha_2^i \\ 1 & a_{3,2}\alpha_3 & \sum_{i=0}^{2-1} a_{3,j}\alpha_3^i & \dots & \sum_{i=0}^{n-1} a_{3,j}\alpha_3^i \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_{n,2}\alpha_n & \sum_{i=0}^{2-1} a_{n,j}\alpha_n^i & \dots & \sum_{i=0}^{n-1} a_{n,j}\alpha_n^i \end{vmatrix} = \begin{vmatrix} 1 & q_1(\alpha_1) & q_1(\alpha_1) & \dots & q_{n-1}(\alpha_1) \\ 1 & q_2(\alpha_2) & q_1(\alpha_2) & \dots & q_{n-1}(\alpha_2) \\ 1 & q_1(\alpha_3) & q_2(\alpha_3) & \dots & q_{n-1}(\alpha_3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & q_1(\alpha_n) & q_2(\alpha_n) & \dots & q_{n-1}(\alpha_n) \end{vmatrix},$$

which give us exactly that $|\hat{V}| = \det\left(q_{j-1}(\lambda_i)\right)_{n\times n}$. □

We still want to express the correlation function as a determinant of a Kernel function. We start by using the fact that $(\det(A))^2 = \det(\sum_{j=1}^n \overline{A_{ji}}A_{jk})$ and Proposition 2.2 to rewrite

$$\prod_{i<j}^n |\lambda_i - \lambda_j|^2 = \det\left(\sum_{l=1}^n \overline{q_{l-1,n}(\lambda_i)}q_{l-1,n}(\lambda_j)\right)_{n\times n}$$

where q_i are orthogonal polynomials with respect to a weight function w. Then, start by putting the weight function inside the determinant in the following manner: Take $w(\lambda_i) = \sqrt{w(\lambda_i)}\sqrt{w(\lambda_i)}$ and multiply the i-th column of the matrix by one of the terms. The other term we multiply by the i-th line of the matrix. Now, we define K_n to be such that

$$\begin{aligned} \mathrm{p}_k(\{\lambda_i\}_{i=1}^k) &= \frac{n!\prod_{l=1}^n r_{n,l}}{\mathcal{Z}_n^Q(n-k)!}\int_{\mathbb{C}^{n-k}} \det\left(\sqrt{w(\lambda_i)w(\lambda_j)}\sum_{l=1}^n \frac{1}{r_{l,n}}\overline{q_{l-1,n}(\lambda_i)}q_{l-1,n}(\lambda_j)\right)_{i,j=1}^n \prod_{i=k+1}^n \mathrm{d}^2\lambda_i, \\ &= \frac{n!\prod_{l=1}^n r_{n,l}}{\mathcal{Z}_n^Q(n-k)!}\int_{\mathbb{C}^{n-k}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^n \prod_{i=k+1}^n \mathrm{d}^2\lambda_i. \end{aligned}$$

At this point it is still not clear why this would be a determinantal point process; we are still to show that the above integral results in a straightforward determinantal expression. To properly use this monic orthogonal polynomials structure, we will prove the fact that the orthogonal kernels $K_n(x,y)$ defined above are known to have a *reproducing property* which allow us to integrate the determinantal structure.

Theorem 2.3. *Let $\{q_{j,n}(x)\}_{j=0}^\infty, \{r_{j,n}\}_{j=0}^\infty$, be as in Definition 2.10 and let $n \in \mathbb{N}$. Consider the orthogonal kernel*

$$K_n(x,y) = \sqrt{w(x)w(y)}\sum_{k=1}^n \frac{1}{r_{k,n}}\overline{q_{k,n}(x)}q_{k,n}(y).$$

Then,

1. The kernel has reproducing properties, i.e.

$$\int_{\mathbb{C}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}^2\lambda_{k+1} = (n-k)\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}.$$

2. The eigenvalue correlations functions of Definition 2.7 are given by

$$p_k(\lambda_1,\lambda_2,\dots,\lambda_n) = \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}.$$

3. The partition function of Definition 2.7 can be expressed as

$$\mathcal{Z}_n^Q = n!\prod_{l=1}^{n} r_{n,l}.$$

Proof. We first note that the expression for the correlation function

$$\mathrm{p}_k(\{\lambda_i\}_{i=1}^k) = \frac{n!\prod_{l=1}^n r_{n,l}}{\mathcal{Z}_n^Q (n-k)!}\int_{\mathbb{C}^{n-k}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}\prod_{i=k+1}^{n}\mathrm{d}^2\lambda_i$$

is such that the j.p.d.f. is

$$p_n(\{\lambda_i\}_{i=1}^n) = \frac{\prod_{l=1}^n r_{n,l}}{\mathcal{Z}_n^Q}\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}$$

and we can calculate the $(n-1)$-th correlation function

$$\begin{aligned}
\mathrm{p}_{n-1}(\{\lambda_i\}_{i=1}^{n-1}) &= \frac{n!\prod_{l=1}^n r_{n,l}}{\mathcal{Z}_n^Q}\int_{\mathbb{C}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}\mathrm{d}^2\lambda_n,\\
&= \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1}\int_{\mathbb{C}} K_n(\lambda_n,\lambda_n)\mathrm{d}^2\lambda_n - \sum_{j=1}^{n-1}\int_{\mathbb{C}} K_n(\lambda_j,\lambda_n)\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1}\mathrm{d}^2\lambda_n\\
&= n\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} - \sum_{j=1}^{n-1}\sum_{\sigma\in S_{n-1}}\operatorname{sgn}(\sigma)\left(\prod_{i\neq j}^{n-1} K_n(\lambda_i,\lambda_{\sigma(i)})\right)\int_{\mathbb{C}} K_n(\lambda_j,\lambda_n)K_n(\lambda_n,\lambda_{\sigma(j)})\mathrm{d}^2\lambda_n,\\
&= n\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} - \sum_{j=1}^{n-1}\sum_{\sigma\in S_{n-1}}\operatorname{sgn}(\sigma)\left(\prod_{i\neq j}^{n-1} K_n(\lambda_i,\lambda_{\sigma(i)})\right)K_n(\lambda_j,\lambda_{\sigma(j)}),\\
&= n\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} - (n-1)\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1},\\
&= (n-(n-1))\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} = \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1}.
\end{aligned}$$

where we used, by its orthogonality properties, that

$$\int_{\mathbb{C}} K_n(\lambda_n,\lambda_n)\mathrm{d}^2\lambda_n = n, \text{ and } \quad \int_{\mathbb{C}} K_n(\lambda_j,\lambda_k)K_n(\lambda_k,\lambda_l)\mathrm{d}^2\lambda_{n-1} = K_n(\lambda_j,\lambda_l).$$

This proves (1). Now, using the above integration recursively for any number k of times, one can also check that (2) is true. Finally, using the above technique again,

$$\mathcal{Z}_n^Q = \prod_{l=1}^{n} r_{n,l}\int_{\mathbb{C}^n}\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}\prod_{i=1}^{n}\mathrm{d}^2\lambda_i = n!\prod_{l=1}^{n} r_{n,l}$$

□

At some point we turn our attention to the univariate eigenvalue density μ as it gives the distribution for a single variable. We note that $\mu = \mathrm{p}_1(\lambda)/n!$ and moreover

$$n!\mathrm{p}_1(\lambda) = \mu(\lambda) = n!\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{1} = K_n(\lambda,\lambda).$$

Therefore, if we understand the behavior of the correlation kernel, we can also state the behavior of the univariate eigenvalue density. Therefore, most results we study of a determinantal point process - such as the asymptotic for μ - follow from properties of the Kernel.

Example 2.9. (Eigenvalue for orthogonal polynomial ensembles) We have shown that for invariant hermitian ensembles the correlation functions are given by

$$\begin{aligned}\mathrm{p}_k(\lambda_1,\lambda_2,\ldots,\lambda_k) &= \frac{n!}{\mathcal{Z}_n^Q (n-k)!}\int_{\mathbb{R}^{n-k}} \prod_{i=1}^{n} \mathrm{e}^{-nQ(\lambda_i)} \prod_{i<j} |\lambda_i-\lambda_j|^2 \prod_{i=k+1}^{n} \mathrm{d}\lambda_i. \\ &= \frac{1}{(n-k)}\int_{\mathbb{R}^{n-k}} \det\left(\sqrt{w(\lambda_i)w(\lambda_j)}\sum_{l=1}^{n}\frac{1}{r_{j,n}} q_{l-1,n}(\lambda_i)q_{l-1,n}(\lambda_j)\right)_{i,j=1}^{n} \prod_{i=k+1}^{n} \mathrm{d}\lambda_i \\ &= \det\left[K_n^Q(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}\end{aligned}$$

where $w(\lambda_i)$ is the chosen weight function. This is the main connection for the theory of random matrices and DPPs - and one that we take major advantage of. On the real line, the orthogonal polynomials $\{q_j(\lambda)\}_{j=0}^{\infty}$ that compose the kernel are known to satisfy the three-term recurrence

$$\lambda q_{j,n}(\lambda) = b_{j-1,n}q_{j-1,n}(\lambda) + a_{k,n}q_{j,n}(\lambda) + b_{k,n}\phi_{k+1,n}(\lambda).$$

where

$$a_{k,n} = \int x q_{k,n}(x)^2 \,\mathrm{e}^{-nQ(x)}\,\mathrm{d}x, \quad b_{k,n} = \int x q_{k,n}(x) q_{k+1,n}(x)\,\mathrm{e}^{-nQ(x)}\,\mathrm{d}x.$$

The polynomials which satisfy the above three-term recurrence for $w(x) = \mathrm{e}^{-nx^2}$ are the Hermite polynomials He_i. We denote the Kernel defined by the Hermite polynomials $K_n^{(H)}$. Moreover, as we are on the real line, we can use the *Christoffel-Darboux* formula

$$K_n(x,y) = \frac{\sqrt{w(x)w(y)}}{r_{n-1,n}} \frac{q_{n,n}(x)q_{n-1,n}(y) - q_{n-1,n}(x)q_{n,n}(y)}{x-y}$$

where $r_{k,n}$ is the norming constant and $q_{k,n}$ is the monic orthogonal polynomial orthogonal with respect to $w(x) = \mathrm{e}^{-nx^2}$. Furthermore, taking the limit $y \to x$ using L'Hopital's rule, for finite n we show that Christoffel-Darboux simplifies to

$$\mu(\lambda) = K_n(\lambda,\lambda) = \frac{w(\lambda)}{r_{n-1}}\left(q_{n-1}(\lambda)\partial_\lambda q_n(\lambda) - q_n(\lambda)\partial_\lambda q_{n-1}(\lambda)\right).$$

and if we have the asymptotic on the n-th and $(n-1)$-th Hermite polynomials, we get the asymptotic for $\mu^{(G)}$.

We now state a few result that will elucidate properties of determinantal point processes and allow us to examine different scales and regimes for the univariate eigenvalue density. Let $f \in C^1(\mathbb{C})$ be a bijection, with continuously differentiable inverse $f^{-1} = g \in C^1(\mathbb{C})$. Then, f maps configurations on $\mathbb{C}$ to configurations on $\mathbb{C}$. The push-forward of a point process $\mathbb{P}$ under such mapping is $f(\mathbb{P})$ and it is again a point process.

Proposition 2.3. *(GIROTTI, 2014) If K is the kernel of a DPP $\mathbb{P}$, then $f(\mathbb{P})$ is also a DPP with kernel*

$$\hat{K}(x,y) = \sqrt{g'(x)g'(y)}K(g(x),g(y))$$

where $f^{-1} = g$.

In particular, if we consider a linear function $f(x) = \alpha(x - x^*)$, for $\alpha > 0$ and $x^* \in \mathbb{C}$, then the transformed point process $f(\mathbb{P})$ is a scaled and translated version of the original point process $\mathbb{P}$. The new correlation kernel is

$$\hat{K}(x,y) = \frac{1}{\alpha}K\left(x^* + \frac{x}{\alpha}, x^* + \frac{y}{\alpha}\right).$$

Example 2.10. (Karlin-McGregor Theorem & Dyson Bridges (KUIJLAARS, 2014)) Consider n independent copies $X_1(t), \cdots, X_n(t)$ of a one-dimensional strong Markov Process with continuous sample paths conditioned so that $X_j(0) = a_j$, where $a_1 < a_2 < \cdots < a_n$ are certain given values. Let $p_t(x,y)$ be some transition probability function of the Markov process. Moreover, let $E_1, E_2, \ldots, E_n$ be Borel sets so that $\sup E_j < \inf E_{j+1}$ for $j = 1, \cdots, n-1$. Then

$$\int_{E_1} \cdots \int_{E_n} \det\left[p_t(a_i, x_j)\right]_{i,j=1}^n \mathrm{d}x_1 \ldots \mathrm{d}x_n$$

is equal to the probability that the paths have not intersected in the time interval $[0,t]$ and $X_j(t) \in E_j$ for $j = 1, \cdots, n$. This is the Karlin-McGregor theorem.

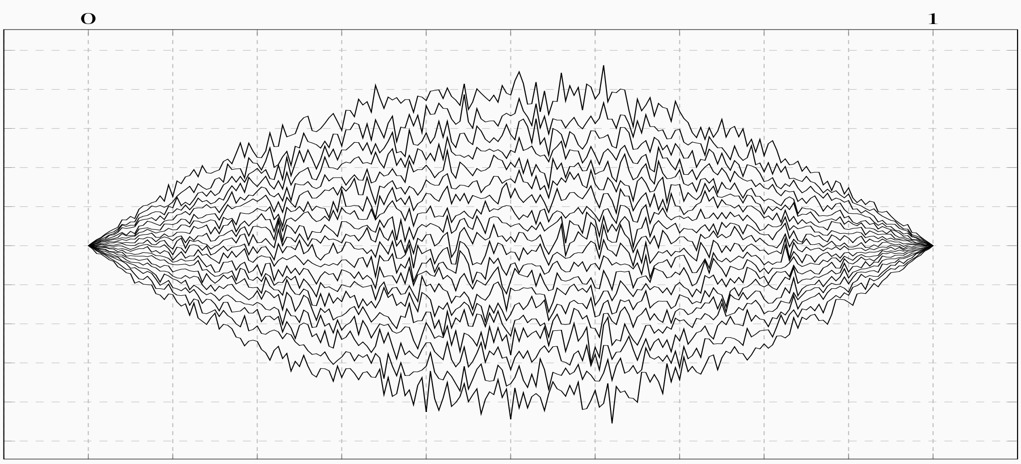

Figure 7 – 20 Non-intersecting Brownian paths for $0 \leq t \leq 1$ s.t. they start and end at $y = 0$.

Take $p_t(a,x) = 1/\sqrt{(2\pi t)}\exp\{-(x-a)^2/2t\}$ a Gaussian transition probability, and consider that we condition our path to not only start at positions $a_1, a_2, \ldots, a_n$ but finish at positions $b_1, b_2, \ldots, b_n$ at some time $T > t$. If we let $a_j \to a$ and $b_j \to b$, the new j.p.d.f. is non-trivially calculated for any t as

$$\begin{aligned} P(x_1,\ldots,x_n) &= \frac{1}{\mathcal{Z}_n}\det\left[p_t(a_i,x_j)\right]_{i,j=1}^n \det\left[p_{T-t}(x_i,b_j)\right]_{i,j=1}^n \\ &= \frac{1}{\mathcal{Z}_n}\det\left[x_j^{i-1}\right]_{i,j=1}^n \prod_{j=1}^n e^{-\frac{x_j^2}{2t}} \det\left[x_j^{i-1}\right]_{i,j=1}^n \prod_{j=1}^n e^{-\frac{x_j^2}{2(T-t)}} \\ &= \frac{1}{\mathcal{Z}_n}\det\left[x_j^{i-1}\right]_{i,j=1}^n \det\left[x_j^{i-1}\right]_{i,j=1}^n \prod_{j=1}^n e^{-\frac{Tx_j^2}{2t(T-t)}} \\ &= \frac{1}{\mathcal{Z}_n}\prod_{1\le i<j\le n}(x_j-x_i)^2 \prod_{j=1}^n e^{-\frac{Tx_j^2}{2t(T-t)}} \end{aligned}$$

Two things are worth noting on this point. The first is that this resembles the j.p.d.f. of the Gaussian Unitary Ensemble! The second thing is that it has a scaling in the weight function based on t. This means that for every t the distribution obeys the same law with a different scaling - it keeps being an determinant process as specified in Proposition 2.3. One can simulate such Brownian motions as below for $T = 1$ as in Figure 7. Notice also on Figure 8 below that the distribution of points is kept the same modulo the re-scaling of $x/((2t(1-t))\sqrt{N})$.

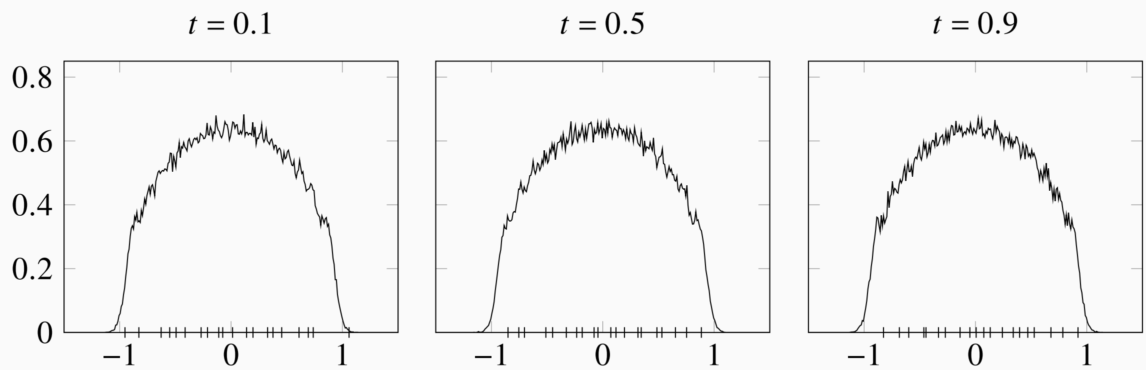

Figure 8 – Three conditions simulated form the 20 Dyson Bridges taken from $t = 0.1, 0.5, 0.9$. For each one the experimental density (line plot) and one sampled position (dashes in axis) were plotted. The three plots are re-scaled by $((2t(1-t))\sqrt{N})$.

We know we can reexamine the Kernels centralized on different points and under different scales. However, so far we only examined finite n Kernels. An essential aspect of RMT, however, is what happens on the thermodynamic limit - when we take $\lim_{n\to\infty} K_n(x,y)$. Therefore, we now consider sequences of finite DPP $\mathbb{P}_n$ with correlation kernel K_n and understand how the limit behaves.

Proposition 2.4. *(GIROTTI, 2014) Let $\mathbb{P}$ and $\mathbb{P}_n$ be determinantal point processes with kernels K and K_n respectively. Let K_n converge point-wise to K*

$$\lim_{n\to\infty} K_n(x,y) = K(x,y)$$

uniformly in x, y over compact subsets of $\mathbb{C}$. Then, the point processes $\mathbb{P}_n$ converges to $\mathbb{P}$ weakly.

The limit K is a kernel that corresponds to a determinantal point process with an infinite number of points. A very important case is that of the limit of an re-scaled and centered determinantal point process. Consider a sequence of kernels $K_n(x,y)$ and a fixed point c of interest. We first perform a centering and re-scaling of the form $x \mapsto Cn^\alpha(x-c)$ for some suitable value of $C, \gamma > 0$. We defined a determinantal point process with the re-scaled kernel

$$\tilde{K}_n(x,y) = \frac{1}{Cn^\alpha} K_n\left(c + \frac{x}{Cn^\alpha}, c + \frac{y}{Cn^\alpha}\right).$$

If $\lim_{n\to\infty} \tilde{K}_n(x,y) = K(x,y)$, K represents the kernel of the infinite DPP in this scaling. Interestingly, in many situations, the same limit K may appear. This phenomenon in known as **universality** in RMT.

Example 2.11. Consider again the Kernel described in Example 2.9 for the Gaussian Unitary Ensemble, $K_n^{(H)}(x,y)$. If

$$\lim_{n\to\infty} \frac{1}{n} K_n^{(H)}(x,x) = \mu_V(x), \quad x \in \mathbb{R},$$

then the asymptotic distribution of the points in the point process (eigenvalues of $\hat{M}_n$) is given by the measure with density μ_V:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f(x_j) = \int f(x)\mu_V(x)\mathrm{d}x.$$

Moreover, suppose μ_V to be supported on one interval $[a,b]$, then for $x^* \in (a,b)$ with $\mu_V(x^*) > 0$, one has

$$\lim_{n\to\infty} \frac{1}{\mu_V(x^*)n} K_n^{(H)}\left(x^* + \frac{x}{\mu_V(x^*)n}, x^* + \frac{y}{\mu_V(x^*)n}\right) = \frac{\sin\pi(x-y)}{\pi(x-y)}.$$

Finally, if near the endpoint b

$$\mu_V(x) = \frac{c}{\pi}(b-x)^{\frac{1}{2}}(1+o(1)), \quad x \to b,$$

then

$$\lim_{n\to\infty} \frac{1}{(cn)^{\frac{2}{3}}} K_n^{(H)}\left(b + \frac{x}{(cn)^{\frac{2}{3}}}, b + \frac{y}{(cn)^{\frac{2}{3}}}\right) = \frac{\mathrm{Ai}(x)\,\mathrm{Ai}'(y) - \mathrm{Ai}'(x)\,\mathrm{Ai}(y)}{x-y},$$

where Ai is the Airy function. Both these limit Kernels are common to all Wigner Matrices.

CHAPTER

3

ASYMPTOTICS

CHAPTER

4

EQUILIBRIUM PROBLEMS

CHAPTER

5

NORMAL MATRIX ENSEMBLES

BIBLIOGRAPHY

BALOGH, F.; GRAVA, T.; MERZI, D. Orthogonal polynomials for a class of measures with discrete rotational symmetries in the complex plane. 2016. Citation on page 35.

BYUN, S.-S.; KANG, N.-G.; SEO, S.-M. Partition functions of determinantal and pfaffian coulomb gases with radially symmetric potentials. **Communications in Mathematical Physics**, Springer Science and Business Media LLC, v. 401, n. 2, p. 1627–1663, Mar. 2023. ISSN 1432-0916. Available: <http://dx.doi.org/10.1007/s00220-023-04673-1>. No citation.

CHAFAï, D.; FERRé, G. Simulating coulomb and log-gases with hybrid monte carlo algorithms. **Journal of Statistical Physics**, Springer Science and Business Media LLC, v. 174, n. 3, p. 692–714, Nov. 2018. ISSN 1572-9613. Available: <http://dx.doi.org/10.1007/s10955-018-2195-6>. Citation on page 35.

DYSON, F. J. The threefold way. algebraic structure of symmetry groups and ensembles in quantum mechanics. **Journal of Mathematical Physics**, v. 3, n. 6, p. 1199–1215, 1962. ISSN 0022-2488. Available: <https://doi.org/10.1063/1.1703863>. Citation on page 26.

EDELMAN, A. **Eigenvalues and condition numbers of random matrices**. Phd Thesis (PhD Thesis) — Massachusetts Institute of Technology, Dept. of Mathematics, 1989. Unpublished. Available: <http://hdl.handle.net/1721.1/14322>. Citation on page 37.

GINIBRE, J. Statistical ensembles of complex, quaternion, and real matrices. **Journal of Mathematical Physics**, v. 6, n. 3, p. 440–449, 03 1965. ISSN 0022-2488. Available: <https://doi.org/10.1063/1.1704292>. Citation on page 27.

GIROTTI, M. **Riemann-Hilbert approach to Gap Probabilities of Determinantal Point Processes**. Phd Thesis (PhD Thesis) — Concordia University, August 2014. Unpublished. Available: <https://spectrum.library.concordia.ca/id/eprint/978910/>. Citations on pages 21, 42, and 44.

HEDENMALM, H.; WENNMAN, A. **Planar orthogonal polynomials and boundary universality in the random normal matrix model**. 2020. Available: <https://arxiv.org/abs/1710.06493>. No citation.

JOHANSSON, K. Course 1 - random matrices and determinantal processes. In: BOVIER, A.; DUNLOP, F.; van Enter, A.; den Hollander, F.; DALIBARD, J. (Ed.). Elsevier, 2006, (Les Houches, v. 83). p. 1–56. Available: <https://www.sciencedirect.com/science/article/pii/S0924809906800387>. Citation on page 21.

KUIJLAARS, A. **Riemann-Hilbert Problems and Multiple Orthogonal Polynomials**. 2014. <http://wis.kuleuven.be/analyse/arno>. [Online; accessed 08-April-2026]. Citation on page 42.

LEBLè, T.; SERFATY, S. Large deviation principle for empirical fields of log and riesz gases. **Invent. Math.**, Springer Science and Business Media LLC, v. 210, n. 3, p. 645–757, dec 2017. No citation.

PIMENTA, J. V. A. **Matrizes aleatórias e simulação de Gases de Coulomb**. 2024. Citation on page 35.

RAHMAN, A. A. **Recursive characterisations of random matrix ensembles and associated combinatorial objects**. 2023. Available: <https://arxiv.org/abs/2301.11428>. Citation on page 21.

SERFOZO, R. F. Chapter 1 point processes. In: **Stochastic Models**. Elsevier, 1990, (Handbooks in Operations Research and Management Science, v. 2). p. 1–93. Available: <https://www.sciencedirect.com/science/article/pii/S0927050705801653>. Citation on page 21.

TREMBLAY, N. **A short introduction to Determinantal Point Processes**. [S.l.]: Institut Henri Poincaré, 2025. Citation on page 36.

WIGNER, E. P. On the distribution of the roots of certain symmetric matrices. **Annals of Mathematics**, v. 67, n. 2, p. 325, mar 1958. Citation on page 24.

ICMC USP
SÃO CARLOS